

Oxygenation During One-Lung Ventilation: The Effects of Inhaled Nitric Oxide and Increasing Levels of Inspired Fraction of Oxygen

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We studied whether inhaled nitric oxide (NO) would improve arterial oxygen tension (P_{aO_2}) and reduce the occurrence of oxygen saturation of hemoglobin (O_2Hb) <90% during one-lung ventilation (OLV). One-hundred-fifty-two patients were ventilated either with or without NO (20 ppm) with an inspired fraction of oxygen (F_{iO_2}) of either 0.3, 0.5, or 1.0 during OLV. Anesthesia was induced and maintained with propofol, remifentanyl, and rocuronium IV, and lung separation was achieved with a double-lumen tube. During OLV, we set positive end-expiratory pressure at 5 cm H_2O , peak pressure at 30 cm H_2O , and end-tidal CO_2 at 30 mm Hg. The nonventilated lung was opened to room air and collapsed. During OLV, three consecutive measurements were performed every

10 min. The operated lung was temporarily ventilated if pulse oximetric saturation (SpO_2) decreased to <91%. SpO_2 <91% occurred in 2 of the 152 patients. SpO_2 overestimated O_2Hb by $2.9\% \pm 0.1\%$. NO failed to improve oxygenation or alter occurrence of O_2Hb <90% during OLV across all time points and all levels of F_{iO_2} . Increasing F_{iO_2} increased oxygenation and decreased occurrence of O_2Hb <90% ($P < 0.001$). At $F_{iO_2} = 1$, P_{aO_2} was higher ($P < 0.01$) and O_2Hb <90% rate tended to be lower ($P = 0.1$) during right versus left lung ventilation. P_{aO_2} was higher in patients undergoing pneumonectomy and lobectomy than in those undergoing metastasectomy or video-assisted operations ($P < 0.05$).

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During one-lung ventilation (OLV) for thoracic surgery, the nonventilated lung remains perfused, leading to an increase in shunt fraction and a decrease in oxygenation. Oxygenation during OLV can be improved by increasing blood flow to the ventilated lung or decreasing blood flow to the nonventilated lung. Inhaled nitric oxide (NO) decreases pulmonary resistance to flow in various animal models and in clinical studies (1,2). Theoretically, the administration of inhaled NO during OLV could increase oxygenation by selectively decreasing pulmonary resistance and increasing blood flow to the ventilated lung. A number of studies have investigated the effects of inhaled NO on oxygenation and pulmonary hemodynamics during OLV (1,3-6). These studies found that during OLV, NO decreases pulmonary artery pressure but fails to increase P_{aO_2} . However, these studies were limited by small patient numbers, coadministration of volatile anesthetics, supine position, closed chest, or a combination of these factors. Furthermore, these studies investigated the effects

of NO on P_{aO_2} but not on desaturation (oxygen saturation of hemoglobin [O_2Hb] <90%) during OLV. We therefore conducted a prospective study to evaluate the effects of NO on P_{aO_2} and on the occurrence of desaturation during OLV. We conducted the investigation at three levels of inspired fraction of oxygen (F_{iO_2}) because changes in P_{aO_2} are best detected during high F_{iO_2} , whereas those in O_2Hb are best seen during low levels of F_{iO_2} . This is because during high F_{iO_2} , even small changes in shunt fraction lead to appreciable changes in P_{aO_2} . During low levels of F_{iO_2} or P_{aO_2} , however, small changes in P_{aO_2} lead to large changes in O_2Hb .

Methods

All patients were anesthetized, ventilated, and monitored according to a standard protocol. Three consecutive measurements of oxygenation and hemodynamics were made at 10, 20, and 30 min after thoracotomy in patients in the lateral decubitus position and after establishment of OLV. During OLV, patients were ventilated either with or without NO. Oxygenation and desaturation were evaluated at an F_{iO_2} of 0.3, 0.5, or 1.0.

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After obtaining approval of the local ethics committee and informed written consent of patients, we studied ASA physical status I-III patients scheduled either for elective video-assisted thoracoscopic surgery (VATS) or open thoracotomy for lung resection in the lateral decubitus position. Exclusion criteria were ASA physical status IV and V and previous lobectomy or bilobectomy in the medical history. Four categories of patients were identified depending on the scheduled surgical procedure: VATS, metastasectomy, lobectomy, or pneumonectomy. Within the type of surgical procedure, patients were randomized on a 1:1 basis to be treated either with or without 20 ppm NO during OLV. Furthermore, patients were ventilated with an Fio_2 of 1.0, 0.5, or 0.3. The levels of Fio_2 were not randomized but used in a subsequent manner ($\text{Fio}_2 = 1$, first group of 50 patients; $\text{Fio}_2 = 0.5$, second group of 52 patients; and $\text{Fio}_2 = 0.3$, last group of 50 patients). Preoperative lung perfusion scans were performed in a standard fashion with $^{99\text{m}}\text{Tc}$ -MAA (macro-aggregated albumin) in 60 of the 152 patients: in 2 patients undergoing VATS, in 20 patients undergoing metastasectomy, in 29 patients undergoing lobectomy, and in 9 patients undergoing pneumonectomy.

Patients were premedicated with 25 mg clorazepate dipotassium PO on the evening before surgery and with 0.5 mg atropine, 7.5-15 mg piritramide, and 2.5 mg droperidol IM 1 h before the scheduled operation. Anesthesia was induced with propofol (1-2 mg/kg) and remifentanyl (0.5-1.0 $\mu\text{g}/\text{kg}$). Rocuronium bromide (0.6 mg/kg) was used to facilitate endotracheal intubation. Anesthesia was maintained with continuous infusion of propofol ($6-10 \text{ mg} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$) and remifentanyl ($5-20 \text{ } \mu\text{g} \cdot \text{kg}^{-1} \cdot \text{h}^{-1}$). We intubated patients with a left- or right-sided double-lumen endotracheal tube (Broncho-CathTM, Mallinckrodt, Athlone, Ireland), size 39F or 41F for men and 37F or 39F for women. The correct position of the tube was controlled with a fiberoptic bronchoscope after intubation and after positioning the patient in the lateral decubitus position (7). If a thoracic epidural catheter was placed (mostly T6-8 interspaces) for postoperative pain relief, the catheter was tested with 3 mL lidocaine 1% and 0.015 mg epinephrine before the induction of general anesthesia. No further epidural medication was given until the end of the study period. A radial arterial line was inserted in every patient. A central venous catheter was inserted only if the condition of the patient and the operative procedure made it necessary. Intraoperative monitoring included electrocardiography, pulse oximetry, nasopharyngeal temperature, neuromuscular monitoring, measurement of ventilation pressures and ventilation volumes, measurement of end-tidal carbon dioxide concentration (Capnomac; Datex, Helsinki, Finland), and urine output. The sensor of the pulse oximeter (Datex, Helsinki, Finland) was placed either on a finger or the ear lobe, whichever led to better Spo_2 readings. To avoid intraoperative hypothermia, an air warming blanket was placed on the

lower extremities, and the arms were covered with a blanket.

An intensive care unit respirator (Evita 2; Dräger, Lübeck, Germany) was used to ventilate the patients during the operative procedure. This was because this ventilator was modified to provide NO at appropriate concentrations and complied with patient safety requirements. The ventilation was pressure controlled, with a peak inspiratory pressure of 30 cm H_2O and a positive end-expiratory pressure of 5 cm H_2O with oxygen or oxygen and air. Ventilation was adjusted to maintain end-tidal CO_2 at approximately 30 mm Hg during OLV. NO (20 ppm) was delivered from a gas cylinder (1000 ppm of NO in nitrogen; AGA, Lidingö, Sweden) into the inspiratory limb of the ventilator by using a delivery device (NODOMO; Dräger, Lübeck, Germany). The NODOMO adjusts NO flow proportionally to the flow of the respirator to maintain a stable NO concentration. In addition, NODOMO can reliably measure NO concentrations in the inspiratory limb of the ventilator.

We started OLV immediately before thoracotomy or trocar placement for thoracoscopy. The operated lung was deflated to atmospheric pressure. Lung collapse was verified by view and monitored by continuous capnometry of the operated lung. After a stabilization period of 10 min, arterial blood gases, including O_2Hb measured by cooximetry (ABL 625; Radiometer Copenhagen, Denmark), heart rate, mean arterial pressure, Spo_2 , and ventilatory variables were measured or noted. Measurements were repeated 20 and 30 min after the beginning of OLV. We planned to discontinue the study if surgical occlusion of blood flow to the nonventilated lung or lung lobe took place; however, this was not necessary in any of the patients studied. If at any time Spo_2 decreased $<91\%$, OLV was interrupted, the collapsed operated lung was reexpanded, and a continuous positive airway pressure of 5-10 cm H_2O with an Fio_2 of 1.0 was applied to maintain oxygenation.

Data were analyzed by using analysis of variance, with the factors time (10, 20, and 30 min), treatment (NO versus no NO), and Fio_2 (0.3, 0.5, and 1.0). A χ^2 test was used for analyzing the frequency of desaturation in the groups of patients studied. Bland and Altman tests were used to assess differences between Spo_2 and O_2Hb during the study. We used the Statistical Package for Social Sciences (SPSS; SPSS Inc., Chicago, IL) version 9.0 for statistical analysis. A $P < 0.05$ was considered significant. Data are presented as mean and SEM or as proportions when appropriate.

Results

We included 152 patients in this study. Demographic data, type of surgery, and preexisting lung or cardiovascular disease were not different among the groups studied (Table 1). There were at least 25 patients in

each of the six groups (NO versus control at $\text{Fio}_2 = 0.3, 0.5$, or 1.0) studied. During OLV, respiratory variables (tidal volume, respiratory frequency, minute volume, and peak and plateau airway pressures) were not different between patients treated with or without NO. Pulse oximetry desaturation, as defined by an $\text{Spo}_2 < 91\%$, occurred in none of the patients ventilated with an Fio_2 of 0.5 or 1.0 and in only two patients ventilated with an Fio_2 of 0.3 .

Mean Pao_2 , Paco_2 , Spo_2 , and O_2Hb did not differ significantly between patients ventilated with or without NO at any Fio_2 level or at any time point (Tables 2–4). Arterial desaturation as defined by $\text{O}_2\text{Hb} < 90\%$ occurred in 20 of 76 patients receiving NO versus 19 of 76 control patients ($P = 0.5$). $\text{O}_2\text{Hb} < 90\%$ occurred in 40 of 219 time points in patients receiving NO versus 36 of 227 time points in control patients ($P = 0.2$). In patients receiving $\text{Fio}_2 = 0.5$, $\text{O}_2\text{Hb} < 90\%$ occurred in 16 of 78 time points in patients receiving NO versus 6 of 78 time points in control patients ($P = 0.04$). In patients receiving $\text{Fio}_2 = 0.3$ or 1.0 , $\text{O}_2\text{Hb} < 90\%$ was not significantly different between the two groups. Levels of methemoglobin ranged between 0.0% and 0.9% (mean $\pm$ SEM, 0.39 ± 0.01) in the Control group and between 0.0% and 1.2% (mean $\pm$ SEM, 0.50 ± 0.01) in patients ventilated with NO ($P < 0.001$).

During OLV, mean Pao_2 , Spo_2 , and O_2Hb increased with increasing Fio_2 (Tables 2 and 3). The frequency of $\text{O}_2\text{Hb} < 90\%$ decreased with increasing Fio_2 (Table 5, $P < 0.001$). $\text{O}_2\text{Hb} < 90\%$ occurred in 26 of 50 patients (50 of 146 time points) with an $\text{Fio}_2 = 0.3$, in 11 of 52 patients (22 of 156 time points) with an $\text{Fio}_2 = 0.5$, and in 2 of 50 patients (5 of 150 time points) with a $\text{Fio}_2 = 1.0$ ($P < 0.01$ for Fio_2 effect for both patients and time points). At $\text{Fio}_2 = 0.3$, there are four missing time points because two patients with an $\text{Fio}_2 = 0.3$ received continuous positive airway pressure in accordance with the study protocol.

Mean difference between Spo_2 and O_2Hb during OLV was $2.9\% \pm 0.1\%$; differences ranged between $+10$ and -4 . The $\text{Spo}_2 - \text{O}_2\text{Hb}$ difference decreased with increasing Fio_2 (3.8 ± 0.2 , 3.0 ± 0.1 , and 2.1 ± 0.2 for $\text{Fio}_2 = 0.3, 0.5$, and 1.0 , respectively; $P < 0.01$).

Pao_2 was significantly higher during right lung ventilation (left thoracic surgery) than during left lung ventilation ($P < 0.01$; Table 6). Across all Fio_2 levels, $\text{O}_2\text{Hb} < 90\%$ occurred in 18 (26%) of 69 patients (39 [19%] of 207 time points) during right lung ventilation and in 23 (28%) of 83 patients (49 [20%] of 245 time points) during left lung ventilation ($P = 0.8$ for both patients and time points). At $\text{Fio}_2 = 1$, $\text{O}_2\text{Hb} < 90\%$ occurred in 5 of 99 time points during left lung ventilation and in 0 of 51 time points during right lung ventilation ($P = 0.1$). At $\text{Fio}_2 = 1.0$, Pao_2 was significantly higher in patients undergoing pneumonectomy (390 ± 26 mm Hg) and in those undergoing lobectomy (256 ± 20 mm Hg) than in those undergoing metastasectomy (164 ± 13 mm Hg) or VATS (166 ± 12 mm Hg) ($P < 0.05$

Table 1. Patient and Operation Characteristics

Variable	Without nitric oxide	With nitric oxide
Age (range)	60 (16–82)	59 (16–78)
Sex (male/female)	55/21	48/28
Cardiovascular disease ^a	28	26
Pulmonary disease ^b	20	23
Operated lung (right/left)	42/34	41/35
ASA status (I/II/III)	7/48/21	8/45/23
Operation		
Metastasectomy	31	32
Lobectomy	17	21
Pneumonectomy	5	4
VATS	22	20

VATS = video-assisted thoracic surgery.

^a Cardiovascular disease included hypertension, coronary artery disease, and valvular heart disease.

^b Pulmonary disease included obstructive or restrictive lung disease or a combination of both.

Table 2. Arterial Oxygen (Pao_2) in Patients Ventilated With or Without 20 ppm Nitric Oxide (NO) at Three Levels of Inspired Fraction of Oxygen (Fio_2)

Fio_2	Time (min)	Pao_2 (mm Hg) ^a	
		With NO	Without NO
0.3	10	68 ± 3	69 ± 3
	20	64 ± 3	67 ± 3
	30	63 ± 3	64 ± 3
0.5	10	93 ± 6^b	101 ± 8^b
	20	90 ± 9^b	96 ± 7^b
	30	84 ± 5^b	94 ± 7^b
1.0	10	181 ± 26^b	224 ± 26^b
	20	205 ± 28^b	185 ± 21^b
	30	211 ± 33^b	198 ± 24^b

^a Mean $\pm$ SEM.

^b Pao_2 increased with increasing Fio_2 . Ventilation with or without NO caused no significant changes.

for pneumonectomy versus all groups and lobectomy versus metastasectomy and VATS). At $\text{Fio}_2 = 1.0$, mean perfusion of the ventilated lung was 45% in patients undergoing metastasectomy (mean of 6 perfusion scans), 53% in patients undergoing lobectomy (mean of 10 perfusion scans), and 68% in patients undergoing pneumonectomy (mean of 4 perfusion scans).

Discussion

We ventilated patients undergoing OLV for lung surgery either with or without 20 ppm NO and found that NO did not improve oxygenation or reduce the occurrence of $\text{O}_2\text{Hb} < 90\%$. In contrast, ventilating patients with increasing levels of Fio_2 during OLV dose-dependently improved oxygenation and reduced the rate of $\text{O}_2\text{Hb} < 90\%$.

Table 3. Oxygen Saturation of Hemoglobin as Measured by Pulse Oximetry (SpO₂) or by Cooximetry (O₂Hb) at Increasing Fraction of Inspired Oxygen (FIO₂) in Patients Ventilated With or Without 20 ppm Nitric Oxide (NO) 10, 20, and 30 min After Thoracotomy and Beginning One-Lung Ventilation

FIO ₂	Time (min)	With NO ^a		Without NO ^a	
		SpO ₂	O ₂ Hb	SpO ₂	O ₂ Hb
0.3	10	95 ± 0.5	92 ± 0.7	94 ± 0.8	91 ± 0.6
	20	95 ± 0.5	91 ± 0.8	95 ± 0.5	91 ± 0.7
	30	95 ± 0.5	90 ± 0.8	95 ± 0.5	91 ± 0.7
0.5	10	97 ± 0.4	94 ± 0.6	97 ± 0.5	95 ± 0.5
	20	97 ± 0.4	93 ± 0.7	97 ± 0.5	94 ± 0.5
	30	98 ± 0.4	93 ± 0.7	97 ± 0.5	94 ± 0.5
1.0	10	98 ± 0.4	96 ± 0.7	99 ± 0.3	97 ± 0.4
	20	98 ± 0.4	96 ± 0.7	98 ± 0.3	97 ± 0.5
	30	98 ± 0.5	96 ± 0.9	98 ± 0.4	96 ± 0.9

^a Mean ± SEM; SpO₂ and O₂Hb (%) increased with increasing FIO₂ ($P < 0.001$). The difference between SpO₂ and O₂Hb decreased with increasing FIO₂ ($P < 0.05$). There was no significant difference in patients treated with or without NO.

Table 4. Minute Ventilation and Arterial CO₂ (Paco₂) Levels at Increasing Fraction of Inspired Oxygen (FIO₂) in Patients Ventilated With or Without 20 ppm Nitric Oxide (NO) 10, 20, and 30 min After Thoracotomy and Beginning One-Lung Ventilation

FIO ₂	Time (min)	With NO ^a		Without NO ^a	
		Minute volume (L/min)	Paco ₂ (mm Hg)	Minute volume (L/min)	Paco ₂ (mm Hg)
0.3	10	7.2 ± 0.3	34 ± 0.6	7.6 ± 0.4	34 ± 0.6
	20	7.6 ± 0.2	32 ± 0.7	7.4 ± 0.4	33 ± 0.6
	30	7.3 ± 0.3	32 ± 0.6	7.6 ± 0.4	33 ± 0.6
0.5	10	7.5 ± 0.4	35 ± 0.5	7.6 ± 0.3	35 ± 0.8
	20	7.4 ± 0.4	33 ± 0.4	7.3 ± 0.2	33 ± 0.8
	30	7.4 ± 0.3	33 ± 0.5	7.3 ± 0.2	33 ± 0.7
1.0	10	7.1 ± 0.4	38 ± 0.9	6.6 ± 0.3	38 ± 1.2
	20	7.2 ± 0.4	36 ± 1.0	7.1 ± 0.3	37 ± 1.3
	30	7.3 ± 0.4	36 ± 0.9	7.1 ± 0.3	37 ± 1.2

^a Mean ± SEM; minute volume did not change significantly among groups, whereas patients ventilated with FIO₂ = 1 had greater mean Paco₂ levels than patients ventilated with FIO₂ <1.0 ($P < 0.001$).

Table 5. Frequency of Desaturation (Arterial Saturation [O₂Hb] <90%) in Patients and During the Three Measurement Time Points During One-Lung Ventilation (OLV) at Increasing Fractions of Inspired Oxygen (FIO₂)

FIO ₂	Desaturation detected ^a (O ₂ Hb <90%)			
	Patients		Time points	
	Yes	No	Yes	No
0.3	26	24	50	96
0.5	11	41	22	134
1.0	2	48	5	145

^a Please note that there were three time points (10, 20, and 30 min after starting OLV) in each patient and that two patients with an FIO₂ of 0.3 were excluded from time point assessment (see Results). Desaturation occurred less frequently with increasing FIO₂ for both patients and time points ($P < 0.01$).

Regarding the effects of NO on Pao₂ during OLV, our study confirms previous findings in smaller patient groups. In a cross-over study, Wilson et al. (6) alternately administered control or study gas (oxygen 70%, N₂O 30% vs oxygen 70%, N₂O 30% with 40 ppm

Table 6. Arterial Oxygen Tension (mm Hg) in Patients with Left or Right Lung Ventilation with Increasing FIO₂ Levels

FIO ₂	Ventilated lung	
	Right	Left
0.3	70 ± 2*	65 ± 2
0.5	103 ± 4*	84 ± 4
1.0	283 ± 18*	169 ± 10

Values expressed as mean ± SEM.

* $P < 0.01$ for right versus left lung ventilation.

NO) every 15 minutes in six patients during OLV. They found that NO did not improve oxygenation, did not decrease pulmonary pressure, and did not decrease shunt fraction during OLV. Moutafis et al. (5) randomized 20 patients in two groups; one group received only oxygen during OLV, and the other group received oxygen and 20 ppm NO. After 30 minutes of OLV, oxygenation was not significantly different in the two groups of patients. In another study,

Fradj et al. (3) allocated 60 patients undergoing lung surgery to a Control or NO (20 ppm) group. After randomization, NO was administered only if P_{aO_2} decreased to <70 mm Hg, which occurred in eight patients of the Control group and eight patients of the NO group. Oxygenation improved in two patients in each group. Thus, our study and previous studies strongly suggest that NO inhalation does not improve oxygenation during OLV. Whereas previous studies were hampered by small patient numbers, coadministration of volatile anesthetics (6), supine position (1), closed chest, or a combination of these factors (3), our study avoids those problems and provides information not only on oxygenation but also on occurrence of hypoxemia ($O_2Hb <90\%$) during OLV.

In our study, inhaled NO failed to reduce the frequency of $O_2Hb <90\%$. Studying the frequency of $O_2Hb <90\%$ may be interesting because at low P_{aO_2} levels, even small changes in P_{aO_2} may lead to appreciable changes in hemoglobin saturation. Surprisingly, in the group receiving $F_{iO_2} = 0.5$, there were significantly more $O_2Hb <90\%$ time points in the NO group as compared with controls. The level of methemoglobin in the NO group does not adequately explain this because mean methemoglobin levels were 0.4 in the Control group and 0.5 in the NO group. Thus, NO does not decrease the rate of $O_2Hb <90\%$ during OLV.

Previous dose-response studies have found that NO levels >5 ppm are effective in producing a response in the pulmonary circulation. In our study, we administered NO at 20 ppm, we monitored NO in the inspiratory limb of the respirator in all patients, and methemoglobin levels were increased in the NO group as compared with the Control group. We used a respirator and an NO delivery device that was produced for the explicit purpose of NO application. Therefore, the negative finding in our study is not the result of technical failure or of small NO levels in inspiratory air.

That inhaled NO during OLV may increase oxygenation by diverting blood from the nonventilated to the ventilated lung is an attractive theory. Furthermore, inhaled NO during OLV may also increase oxygenation by improving ventilation-perfusion matching within the ventilated lung. Apparently, these mechanisms were either not operative or insufficient to increase oxygenation or reduce the occurrence of $O_2Hb <90\%$ during OLV. A number of reasons may explain this. During OLV in the lateral decubitus position, approximately 75%–80% of the cardiac output flows through the ventilated, dependent lung. This suggests that pulmonary vessels in the ventilated lung are dilated to accommodate this increased flow, and further dilation with NO is not possible. Studies investigating the effects of NO in patients with lung injury support this hypothesis. In a clinical trial, Benzing et al. (8)

studied whether cardiac output may modify the effects of NO on venous admixture in patients with lung injury. They found that in patients with high cardiac output, NO was less effective in decreasing venous admixture than in patients with low cardiac output. Thus, increased flow in the ventilated, dependent lung during OLV may have prevented NO from further increasing flow or improving ventilation-perfusion matching.

We had decided to ventilate the collapsed lung only if S_{pO_2} decreased to $<91\%$. Surprisingly, S_{pO_2} did not decrease $<91\%$ in any of the 102 patients ventilated with an F_{iO_2} of 0.5 or 1.0, and in only 2 of 50 patients ventilated with an $F_{iO_2} = 0.3$. Because S_{pO_2} overestimated O_2Hb , we used $O_2Hb <90\%$ as a measure of hypoxemia for the purpose of this study. Our study shows that during OLV, patients ventilated with increasing F_{iO_2} levels show corresponding decreases in occurrence of $O_2Hb <90\%$. This means that increasing F_{iO_2} levels may be an effective means of increasing oxygenation or treating or preventing hypoxemia during OLV. The use of high levels of F_{iO_2} is detrimental in animal models of lung injury (9) and promotes atelectasis during anesthesia (10). Whether the administration of high levels of oxygen during the relatively short period of OLV may be injurious seems unlikely. However, other maneuvers, such as application of continuous positive airway pressure to the nonventilated lungs, may obviate the need for high F_{iO_2} during OLV.

Despite an F_{iO_2} of 1.0, $O_2Hb <90\%$ occurred at least once in 2 (4%) of 50 patients or in 5 ($<4\%$) of 150 time points. This finding means that hypoxia cannot be completely eliminated when using $F_{iO_2} = 1.0$ but that it occurs very infrequently. It should be noted that we used O_2Hb to define hypoxemia and desaturation. O_2Hb is the measured arterial saturation devoid of carboxyhemoglobin and methemoglobin. Previous studies using S_{aO_2} (arterial saturation, which may include carboxyhemoglobin and methemoglobin) and S_{pO_2} have reported hypoxemia rates of up to 10% during OLV (11–13). Lower levels in our study may be due to increased monitoring of double-lumen tube position at our center (7) and the ventilation strategy used during OLV. Because of study circumstances (see Methods), we used an intensive care unit ventilator and a pressure-controlled mode to ventilate patients during OLV. Most other studies reporting on oxygenation have used volume-controlled ventilation. Tugrul et al. (11) have shown that pressure-controlled ventilation may lead to higher levels of P_{aO_2} than volume controlled ventilation.

The two patients with $O_2Hb <90\%$ during $F_{iO_2} = 1.0$ had their left lung ventilated during OLV. Because the left lung is smaller than the right lung, oxygenation is usually lower during left lung ventilation as

compared with right lung ventilation. The small difference in shunt volume between right and left lung ventilation leads to a difference in P_{aO_2} that increases with increasing F_{IO_2} levels. At $F_{IO_2} = 1.0$, mean P_{aO_2} was 283 mm Hg during right lung ventilation and 169 mm Hg during left lung ventilation. Furthermore, occurrence of $O_2Hb < 90\%$ tended to be less ($P = 0.1$) during right versus left lung ventilation. In agreement with our finding, Slinger et al. (13) found the side of surgery to be an independent predictor of P_{aO_2} during OLV; however, they report neither P_{aO_2} values nor desaturation rates.

Our study also shows that oxygenation was better during pneumonectomy and lobectomy as compared with metastasectomy and thoracoscopy. This may be explained, in part, by lung perfusion studies. In our study and at $F_{IO_2} = 1.0$, the perfusion of the ventilated lung was higher during lobectomy (55%) and pneumonectomy (68%) as compared with metastasectomy (45%). This means that the pathology involved in patients undergoing lobectomy and pneumonectomy may increasingly impair the perfusion of the nonventilated lung, thus decreasing shunt fraction and improving oxygenation. Thus, oxygenation during OLV may also depend on whether the right or left lung is ventilated and whether blood flow to the operated lung is impaired.

In conclusion, our study shows that NO administration is not associated with better oxygenation during OLV. In contrast, increasing F_{IO_2} levels were associated with better oxygenation during OLV.

References

1. Rich GF, Lowson SM, Johns RA, et al. Inhaled nitric oxide selectively decreases pulmonary vascular resistance without impairing oxygenation during one-lung ventilation in patients undergoing cardiac surgery. *Anesthesiology* 1994;80:57-62.
2. Steudel W, Hurford WM, Zapol WM. Inhaled nitric oxide: basic biology and clinical applications. *Anesthesiology* 2000;91:1090-121.
3. Fradj K, Samain E, Delefosse D, et al. Placebo-controlled study of inhaled nitric oxide to treat hypoxaemia during one-lung ventilation. *Br J Anaesth* 1999;82:208-12.
4. Ismail-Zade IA, Vuylsteke A, Ghosh S, et al. Inhaled nitric oxide and one-lung ventilation in the lateral decubitus position [letter]. *J Cardiothorac Vasc Anesth* 1997;11:926-7.
5. Moutafis M, Liu N, Dalibon N, et al. The effects of inhaled nitric oxide and its combination with intravenous almitrine on P_{aO_2} during one-lung ventilation in patients undergoing thoracoscopic procedures. *Anesth Analg* 1997;85:1130-5.
6. Wilson WC, Kapelanski DP, Benumof JL, et al. Inhaled nitric oxide (40 ppm) during one-lung ventilation, in the lateral decubitus position, does not decrease pulmonary vascular resistance or improve oxygenation in normal patients. *J Cardiothorac Vasc Anesth* 1997;11:172-6.
7. Klein U, Karzai W, Wohlfart M, et al. Role of fiberoptic bronchoscopy in conjunction with the use of double-lumen tubes for thoracic anesthesia: a prospective study. *Anesthesiology* 1998;88:346-50.
8. Benzing A, Loop T, Mols G, Geiger K. Effect of inhaled nitric oxide on venous admixture depends on cardiac output in patients with acute lung injury and acute respiratory distress syndrome. *Acta Anaesthesiol Scand* 1996;40:466-74.
9. Erdmann AJ, Huttmeier PC, Landolt C, Zapol WM. Pure oxygen breathing increases sheep lung microvascular permeability. *Anesthesiology* 1983;58:153-8.
10. Rothen HU, Sporre B, Engberg G, et al. Influence of gas composition on recurrence of atelectasis after a reexpansion maneuver during general anesthesia. *Anesthesiology* 1995;82:832-42.
11. Tugrul M, Camci E, Kardeniz H, et al. Comparison of volume controlled and pressure controlled ventilation during one-lung anaesthesia. *Br J Anaesth* 1997;79:306-10.
12. Hurford WM, Alfille PH. A quality improvement study of the placement and complications of double-lumen endobronchial tubes. *J Cardiothorac Vasc Anesth* 1993;7:517-20.
13. Slinger P, Suissa S, Triolet W. Predicting arterial oxygenation during one-lung anaesthesia. *Can J Anaesth* 1992;39:1030-5.